

A
PROJECT REPPORT
ON
EcoTrace:Carbon Footprint Accounting

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering

In

Information Technology

by

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University of Mumbai

2025-26

CERTIFICATE

This is to certify that the project entitled **“EcoTrace:Carbon Footprint Accounting”** is a bonafide work of **“Bhakti Ambrale(01), Ayush Chaurasia(06), Shravani Sarang(50), Karan Yadav(64) ”** submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of **“Bachelor of Engineering”** in **“Information Technology”**.

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Project Report Approval for B.E.

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Examiners

1.-----

2.-----

Date:

Place:

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signature

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TABLE OF CONTENT

Sr.No.	Topic	Page No.
	Certificate	i
	Approval Sheet	ii
	Declaration	iii
	Acknowledgement	iv
	List of Figures	vi
	List of Table	vii
	Abstract	viii
1.	Introduction	1
2.	Literature Survey	
3.	Proposed Work	
	3.1 Requirement Analysis	
	3.1.1 Scope	
	3.1.2 Feasibility Study	
	3.1.3 Hardware & Software Requirement	
	3.2 Problem Statement	
	3.3 Project Design.....	
	3.3.1 Proposed Algorithm	
	3.3.2 Diagrams.....	
	3.4 Methodology.....	
	3.4.1 Green IT Practices implemented.....	
	3.5 Implementation Plan	
4.	Testing	
	4.1 Performance Metrics.....	
	4.2 Sustainability Software Metrics.....	
	4.3 Test Cases.....	
5.	Results and Discussion	

6. Conclusion and Future Scope.....

References

Publications by the candidate.....

LIST OF FIGURES

Figure No.	Name of the Figure	Page No.

LIST OF TABLES

Table No.	Name of the Table	Page No.

ABSTRACT

Climate change is one of the most pressing challenges of our time, driven largely by greenhouse gas (GHG) emissions from human activities. To mitigate its impact, accurate carbon accounting has become essential for businesses worldwide. However, for Micro, Small, and Medium Enterprises (MSMEs), carbon accounting is often complex, expensive, and time-consuming. Most existing tools are designed for large corporations and require specialized expertise, leaving MSMEs and Academic Institutes without affordable and practical solutions. Our project, EcoTrace, addresses this gap by providing a simple, AI-powered carbon accounting tool. The system leverages modern technologies such as Optical Character Recognition (OCR), Natural Language Processing (NLP), and Machine Learning to automate the collection, translation, and analysis of carbon-related data. EcoTrace transforms raw, unstructured data like utility bills, invoices, or receipts into meaningful insights, actionable recommendations, and compliance-ready reports. Unlike traditional reporting tools, EcoTrace emphasizes usability, affordability, and business value, making sustainability accessible to companies that may not have in-house expertise.

1. Introduction

Climate change poses one of the greatest threats to humanity, ecosystems, and economies in the 21st century. Rising global temperatures, extreme weather events, and resource depletion are closely linked to the accumulation of greenhouse gases (GHGs), primarily driven by human activities such as energy consumption, transportation, and industrial processes. To combat this crisis, carbon accounting and sustainability reporting have emerged as critical tools for monitoring and reducing emissions at all scales—from global industries to local institutions.

In this project, we focus on building a carbon footprint accounting system for our college campus. Educational institutions, though smaller in scale compared to industries, still contribute significantly to carbon emissions through electricity consumption, water usage, transportation, food services, and daily operational activities. Despite this impact, systematic carbon accounting practices are often missing in academic environments.

Traditional carbon accounting tools are typically resource-intensive and designed for corporations, making them unsuitable for campus-level adoption. Our solution addresses this gap by developing a simple, user-friendly, and automated system tailored to the needs of a college environment. The system integrates technologies such as Optical Character Recognition (OCR), Natural Language Processing (NLP), and Machine Learning (ML) to extract and analyze data from utility bills, invoices, and activity logs. This automation reduces the manual effort required, minimizes errors, and provides accurate insights into the institution's carbon footprint.

Through this project, we aim to create a practical tool that not only tracks and reports emissions but also empowers our college community to make informed decisions about energy efficiency, resource optimization, and sustainable practices. By embedding carbon accounting into the academic ecosystem, our college can actively contribute to climate change mitigation while fostering a culture of environmental responsibility among students and staff.

2. Literature Survey

Year	Title	Objective	Methodology	Challenges
2025	Enhancing Supply Chain Sustainability Using AI for Carbon Footprint Analysis and Optimization [1]	Explore how AI can enhance supply chain sustainability through carbon footprint analysis and optimization.	Literature review, application of AI models (data collection, emission estimation), and case studies.	Data inconsistencies, legacy system integration, real-time data acquisition, model accuracy, and scalability.
2025	AI-Powered Carbon Accounting: Redefining ESG Standards for Corporate Sustainability[2]	Explore how AI can transform carbon accounting and ESG reporting by automating tracking, improving accuracy, and ensuring compliance.	AI integration (ML, NLP) in carbon accounting and ESG reporting.	Data privacy/security, algorithmic bias, infrastructure dependency, regulatory fragmentation, governance needs, and high costs.
2025	Sustainable and Inclusive Low-Carbon Transport Scenarios for Surat City, India[3]	Examine climate mitigation and SDG delivery in Surat's transport sector using scenario analysis.	Scenario development (Base, CMP, Deep Decarbonization, SDG), GHG emissions modeling (ASIF), and stakeholder consultations.	Feasibility of ambitious scenarios, trade-offs (decarbonization vs. social equity), infrastructure scalability, and behavioral change.
2025	A review and development of an enhanced carbon footprint calculator to facilitate knowledge of individual carbon emissions.[4]	Create and validate an improved, user-friendly carbon calculator to educate users and encourage action.	Reviewed 1,500+ calculators, developed a new tool (LT V1/V2), and tested it with 100+ participants.	Many calculators aren't user-friendly; difficulty translating user knowledge into real-world action.

Year	Title	Objective	Methodology	Challenges
2025	Carbon Footprint of Electricity Transmission: Insights from a Brazilian Case Study	Quantify the carbon footprint of electricity transmission via a Brazilian case study.	Transmission scenario analysis (ANAREDE software) and life cycle inventory analysis (construction, operation, decommissioning).	Dependence on secondary data, accessing technical info, lack of dynamic modelling, and limited exploration of new tech.
2024	Environmental Footprint Impacts of Green Energies, Green Energy Finance and Green Governance in G7 Countries	Assess the impact of green energy, finance, and governance on ecological/carbon footprints in G7 countries (1990-2020).	Panel data analysis (1990-2020), econometric modeling (FMOLS, DOLS), and causality testing.	Conflicting findings on economic impacts, limited scope (institutional/social factors), and exclusion of urbanization factors.
2024	Carbon Footprint Management in Global Supply Chains: A Data-Driven Approach Utilizing Artificial Intelligence Algorithms	Apply AI to reduce carbon emissions in global supply chains through optimization and predictive analysis.	Used Random Search for optimization and a CNN for emission prediction..	Data quality/availability, computational complexity, lack of AI transparency, and real-time processing limits.
2023	Twelve Years Research Journey of Carbon Accounting[8]	Systematically review carbon accounting research (2011–2022) using bibliometric analysis.	Bibliometric and content analysis (NVivo) of 62 Scopus-indexed articles.	Lack of standardized methodologies, limited Scope 3 tracking, inconsistent regulations, and data integration gaps.

Year	Title	Objective	Methodology	Challenges
2022	Carbon Accounting: A Systematic Literature Review and Directions for Future Research[9]	Review carbon accounting research to identify gaps and future directions.	Systematic literature review (27 articles) with thematic analysis.	Lack of transparency/reliability, weak link between GHG inventories and decision-making, and insufficient policy guidance.
2020	Research on Extended Carbon Emissions Accounting Method and Its Application in Sustainable Manufacturing[10]	Develop an Extended Carbon-Emission Accounting (ECEA) method that includes labor and capital emissions..	Combined extended exergy accounting with carbon footprint calculation; applied to compare manufacturing processes.	Exclusion of labor/capital factors, difficulty quantifying socio-economic emissions, and lack of standard conversion factors.
2020	Towards the Systematic Reporting of the Energy and Carbon Footprints of Machine Learning[11]	Introduce a framework to track energy use and carbon emissions in ML research.	Developed an "experiment-impact-tracker" framework with real-time energy monitoring and case studies.	Measurement complexity, accuracy of proxy metrics, regional energy grid variations, and reproducibility issues.

3. Proposed Work

3.1. Requirement Analysis

3.1.1. Scope

The scope of EcoTrace is to provide a lightweight, intelligent, and affordable carbon accounting solution for Academic Institute, simplifying the measurement, tracking, and reporting of greenhouse gas (GHG) emissions. The system automates data collection from invoices, utility bills, and receipts using OCR, interprets textual data with NLP to standardize emission factors, estimates and categorizes emissions through machine learning models, generates compliance-ready reports, provides actionable recommendations to reduce emissions, and presents all information via a user-friendly dashboard designed for non-expert users.

3.1.2. Feasibility Study

A feasibility study assesses the practicality and viability of a project before its development. For EcoTrace, the feasibility can be evaluated across three main dimensions: technical, economic, and operational feasibility.

1. Technical Feasibility

EcoTrace relies on well-established technologies such as OCR (Tesseract), and web development frameworks like Flask/React. These technologies are open-source, widely supported, and suitable for small to medium-sized applications. The project does not require highly complex infrastructure, and the integration of AI modules for data extraction and analysis is achievable within the available skill set and timeline. Therefore, the technical feasibility is high.

2. Economic Feasibility

The project leverages free or low-cost tools, libraries for backend operations. By minimizing software and hardware costs, Eco Trace remains affordable to develop and deploy. Additionally, MSMEs can benefit from reduced labour costs associated with manual carbon accounting. Overall, the economic feasibility is strong, making the project cost-effective for both developers and end-users.

3. Operational Feasibility

EcoTrace is designed for non-expert users, with a simple dashboard, automated data processing, and actionable insights. It aligns with the operational needs of MSMEs, which often lack sustainability expertise. The tool's usability ensures that businesses can adopt it without extensive training or process changes, confirming high operational feasibility.

3.1.3 Hardware And Software Requirements

Hardware Requirements

- Processor: Intel i7 or higher recommended for faster AI processing.
- RAM: Minimum 8 GB, recommended 16 GB for smoother handling of large datasets and OCR processing.
- Storage: Minimum 256 GB SSD for faster read/write operations and storing data files.

Software Requirements

- Operating System: Windows 11
- Programming Languages & Libraries:
 - Python 3.x – core programming language.
 - OCR: Tesseract OCR with pytesseract.
 - NLP: spacy or NLTK for text processing.
 - Image Processing: OpenCV or Pillow.
- Web Frameworks: Flask (backend), React.js (frontend).
- Database: SQLite for lightweight storage
- Visualization Libraries: Plotly.js, Chart.js, or D3.js for interactive dashboards.
- Others: Git for version control, Node.js & npm for frontend package management.

3.2 Problem Statement

Climate change is accelerating due to increasing greenhouse gas emissions, yet individuals and small institutions lack accessible tools to accurately measure, track, and reduce their carbon footprint. Most existing carbon accounting systems are complex, expensive, and designed for large organizations, making them unsuitable for students and small-scale users.

Additionally, everyday carbon-generating activities such as electricity consumption, fuel usage, and resource utilization are recorded in unstructured formats like utility bills, invoices, and PDFs. Extracting meaningful data (e.g., units consumed, dates, and usage values) from these documents manually is time-consuming and prone to errors.

There is a need for an intelligent, cost-effective, and user-friendly system that can Automatically extract consumption data from bills and documents, Calculate carbon emissions accurately, Provide personalized recommendations to reduce emissions, Increase carbon awareness among students and individuals

3.3. Project Design

3.3.1. Proposed Algorithm

EcoTrace presents a systematic approach to designing, developing, and implementing an AI-powered carbon accounting tool. The process is divided into five key stages:

1. Data Collection and Ingestion

- Objective: Gather all relevant carbon-related data from multiple sources.
- Approach: Utilize Optical Character Recognition (OCR) with libraries such as Tesseract and OpenCV/Pillow to extract text from invoices, receipts, utility bills, and PDFs. This automates data entry and improves accuracy.
- Outcome: Structured digital data ready for further processing.

2. Data Translation and Standardization

- Objective: Convert collected data into a standardized format suitable for carbon calculation.
- Approach: Apply Natural Language Processing (NLP) using spaCy or NLTK to interpret textual information (e.g., fuel type, electricity usage, travel distance) and map it to emission factors. Units are normalized to a standard metric (kg CO₂-equivalent).
- Outcome: Clean, standardized, and consistent data suitable for analysis.

3. Carbon Emission Calculation

Objective:

To accurately quantify the carbon footprint generated from various consumption activities such as electricity usage and fuel consumption.

Approach:

- Use standardized emission factors aligned with recognized frameworks such as the GHG Protocol.
- Perform calculations using Python libraries like Pandas and NumPy for efficient numerical computation and data aggregation.
- Apply deterministic mathematical formulas to compute emissions:

$$\text{Carbon Emission} = \text{Units (kWh)} \times \text{Emission Factor}$$

- Categorize emissions into:
 - Scope 1: Direct emissions (e.g., petrol, diesel, LPG)
 - Scope 2: Indirect emissions from purchased electricity
 - Partial Scope 3: Other indirect emissions
- Aggregate emissions on a monthly and yearly basis for reporting and analysis.

Outcome:

Accurate and standardized carbon emission values categorized by emission scope, enabling reliable tracking, comparison, and sustainability assessment

4. Analysis and Identification of Hotspots

- Objective: Identify key contributors to the organization's carbon footprint.
- Approach: Analyze data using statistical techniques and visualizations (e.g., Plotly.js, Chart.js, D3.js) to detect patterns and carbon "hotspots."
- Outcome: Insights into high-emission activities or processes, enabling targeted mitigation strategies.

5. Actionable Recommendations and Reporting

- Objective: Provide practical guidance and generate compliance-ready reports.
- Approach: Implement a recommendation engine using rule-based systems or collaborative filtering to suggest emission reduction strategies. Automated report generation ensures alignment with frameworks such as the GHG Protocol.
- Outcome: Clear, actionable recommendations and ready-to-submit sustainability reports.

3.3.2Diagrams



Fig.1.Flowchart

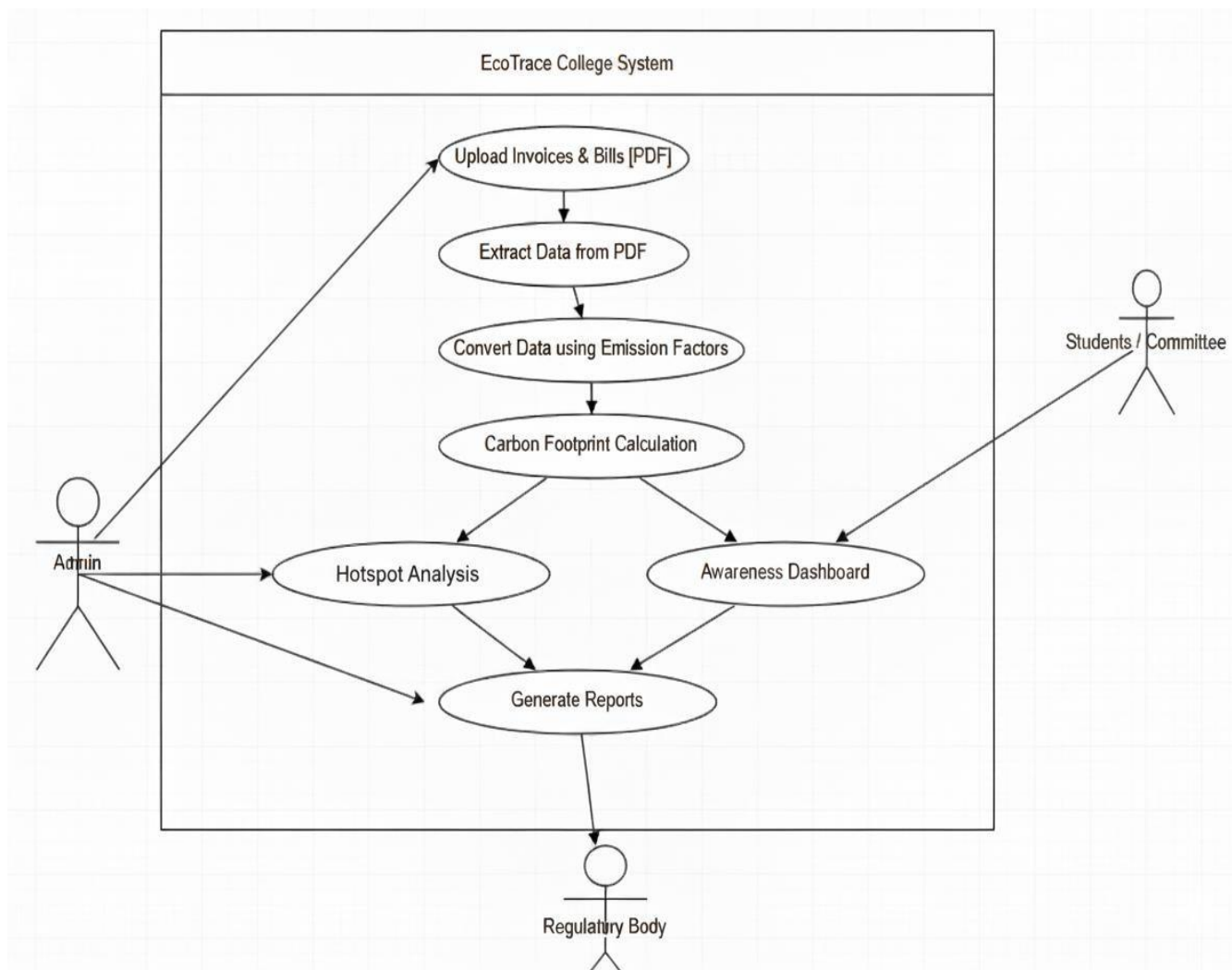


Fig.2.Use Case Diagram

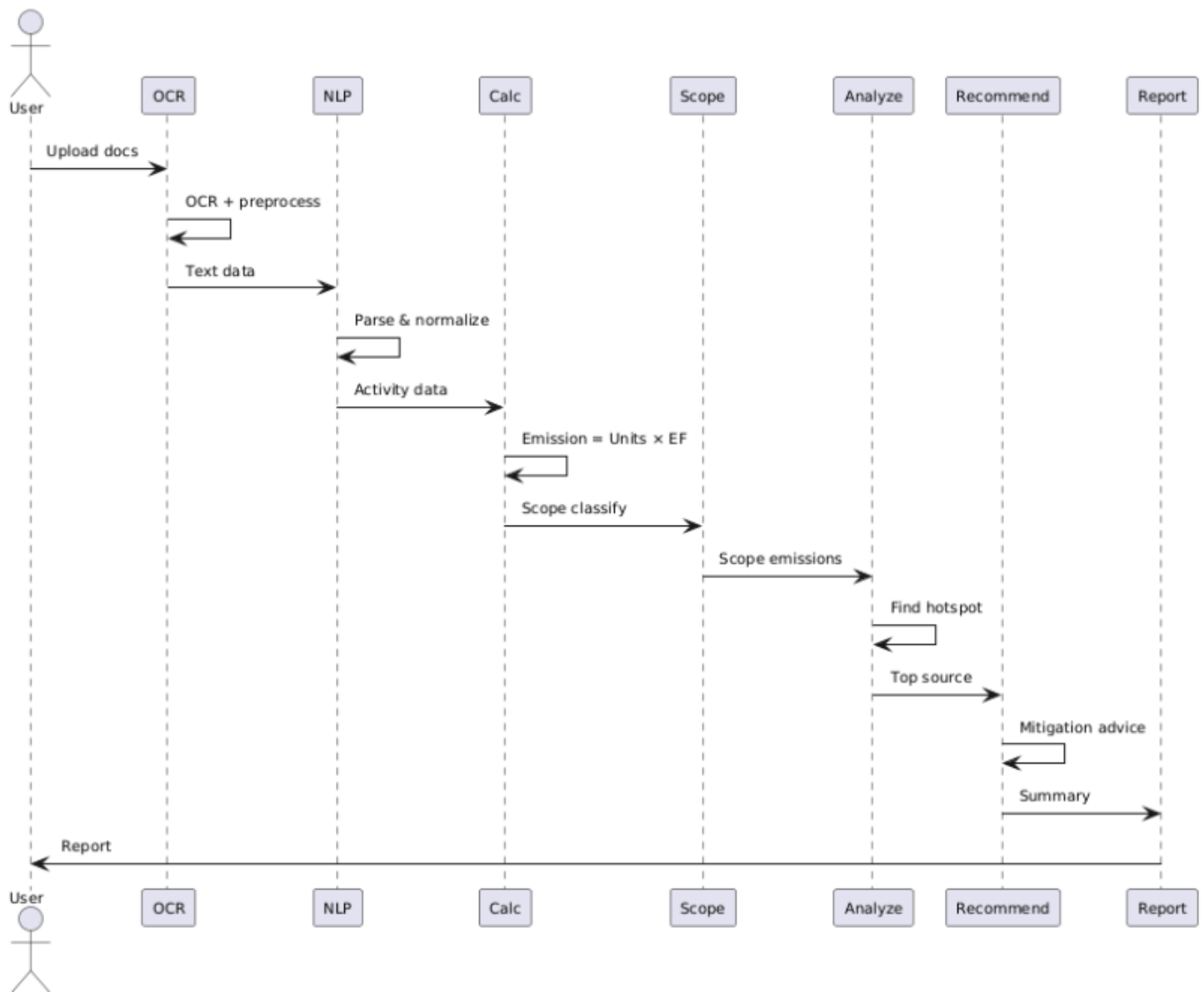


Fig.3.UML Sequence Diagram

3.4. Methodology

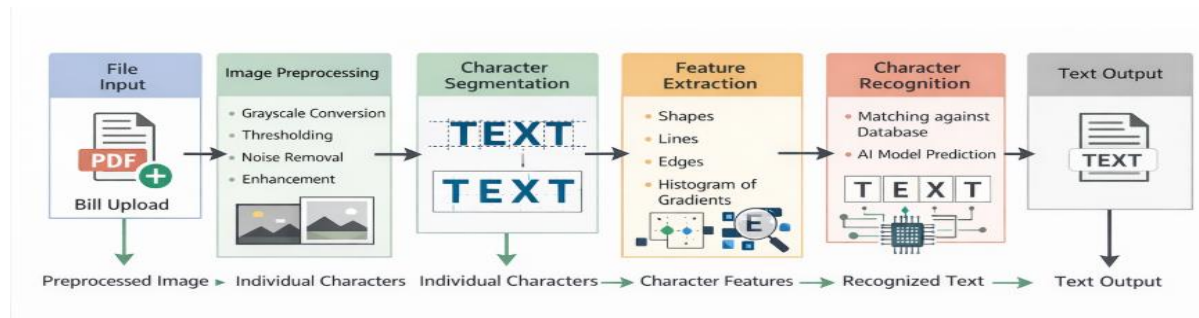
The proposed Carbon Footprint Accounting System adopts a modular, data-driven design that integrates document intelligence, analytical processing, and generative AI to automate carbon emission measurement and decision support. The system focuses on accuracy, scalability, and compliance with sustainability reporting standards. This section describes the system architecture, data acquisition pipeline, processing techniques, communication workflow, and evaluation strategy used during implementation.

A. System Architecture

The system architecture is structured into four interconnected layers that operate sequentially to convert raw activity data into actionable sustainability insights:

- **Data Ingestion Layer:** Automates extraction of carbon-related information from invoices, bills, and documents using OCR.
- **Processing and Standardization Layer:** Cleans, interprets, and normalizes extracted data into structured emission-ready formats.
- **Carbon Calculation and Analysis Layer:** Computes emissions and identifies high-impact sources.
- **Recommendation and Reporting Layer:** Generates AI-driven mitigation strategies and summarized reports.

All layers communicate through a centralized backend, ensuring smooth data flow and real-time processing.



B. Data Collection and Ingestion

This layer is responsible for converting physical and digital documents into machine-readable data.

- **Input Sources:** Utility bills, fuel invoices, travel receipts, and PDF documents.
- **OCR Engine:** Tesseract OCR integrated with OpenCV and Pillow for preprocessing tasks such as grayscale conversion, noise removal, and thresholding.
- **Output:** Extracted raw textual data representing energy consumption, fuel usage, and activity metrics.

This automation minimizes manual entry errors and improves data acquisition efficiency.

C. Data Translation and Standardization

Extracted textual data is transformed into a consistent format.

- **Text Parsing:** Rule-based processing and lightweight NLP techniques are used to identify activity types, units, and numerical values.
- **Unit Normalization:** All measurements are converted into standardized units (e.g., kWh, liters, kilometers)
- **Emission Mapping:** Activities are linked to predefined emission factors aligned with standard carbon accounting guidelines.

The output of this stage is clean, structured data expressed in kg CO₂-equivalent.

D. Carbon Emission Calculation

The standardized data is used to compute carbon emissions accurately.

- **Computation Framework:** Python-based processing using Pandas and NumPy.
- **Emission Categorization:** Calculations are grouped into Scope 1, Scope 2, and partial Scope 3 emissions.

This module acts as the core “carbon calculator” of the system.

E. Analysis, Dominant Emission Hotspot Identification, and Targeted Recommendation

After carbon emission computation, the system performs an aggregate analysis to determine the single largest contributing source to the overall carbon footprint.

- **Dominant Hotspot Identification:**

Emissions from all recorded activities are aggregated and ranked. The activity with the maximum CO₂-equivalent contribution is identified as the primary emission hotspot.

- **Focused Hotspot Analysis:**

Instead of distributing attention across multiple minor sources, the system concentrates on the dominant emission source to maximize reduction impact. Parameters such as activity type, consumption level, and emission intensity are evaluated.

- **AI-Driven Recommendation for Dominant Source:**

Only the identified dominant hotspot is provided as input to the API. Based on the hotspot context and emission magnitude, generates specific and practical emission reduction recommendations tailored to that single high-impact source.

- **Outcome:**

The system delivers one prioritized recommendation set aligned with the largest emission contributor, enabling organizations to focus mitigation efforts where they yield the highest benefit.

3.4.1. Green IT Practices implemented

The EcoTrace system incorporates multiple Green IT practices to minimize environmental impact while ensuring efficient and sustainable system operation.

1. Paperless Digital Processing

- Physical bills, invoices, and receipts are digitized using OCR.
- Eliminates manual paperwork and reduces paper consumption.
- Supports long-term digital storage and retrieval.

2. Energy-Efficient Computing

- Lightweight Python libraries (Pandas, NumPy) are used for optimized computation.
- Avoids unnecessary heavy processing, reducing CPU and memory usage.
- Batch processing of data minimizes repeated computation cycles.

3. Optimized Data Storage

- Stores only structured and relevant emission data.
- Reduces redundant data storage through normalization and aggregation.
- Aggregated monthly and yearly data lowers storage and processing overhead.

4. Targeted Emission Analysis

- Focuses on identifying the single dominant emission hotspot.
- Prevents unnecessary analysis of low-impact sources.
- Enables efficient decision-making with minimal computational cost.

5. AI-Assisted Decision Support

- AI recommendations are generated only for the highest emission source.
- Reduces excessive API calls and computational load.
- Promotes efficient use of AI resources.

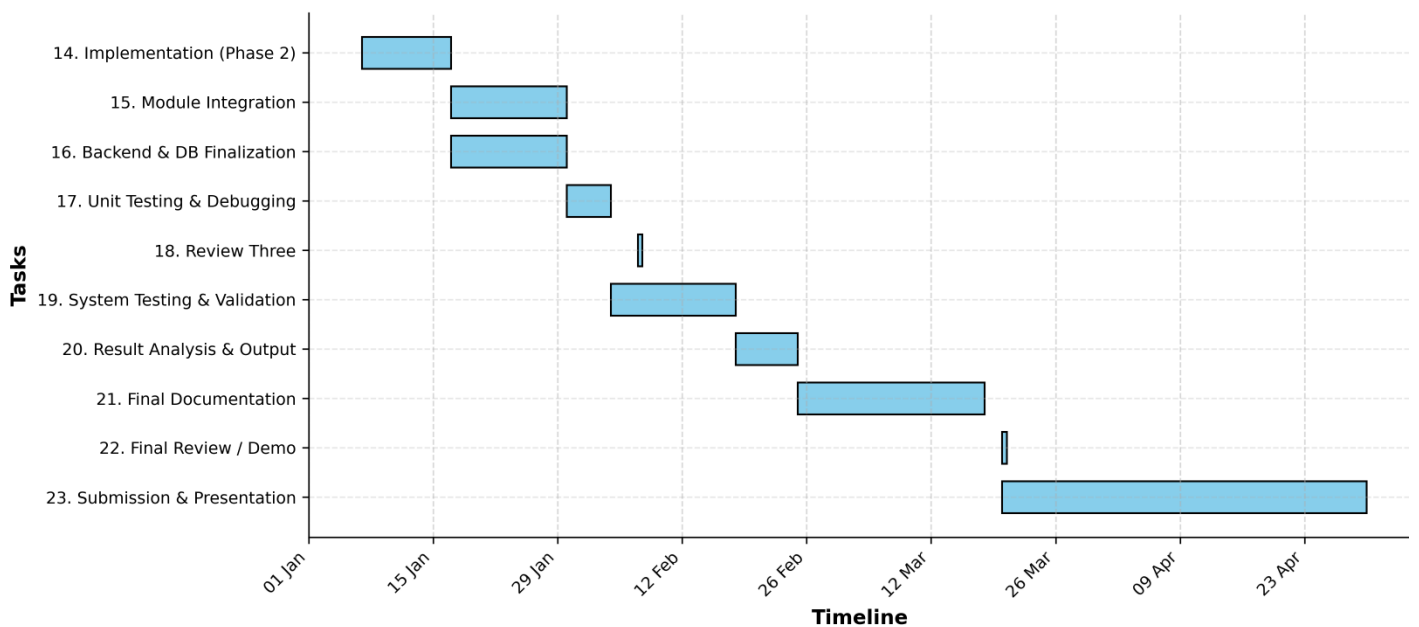
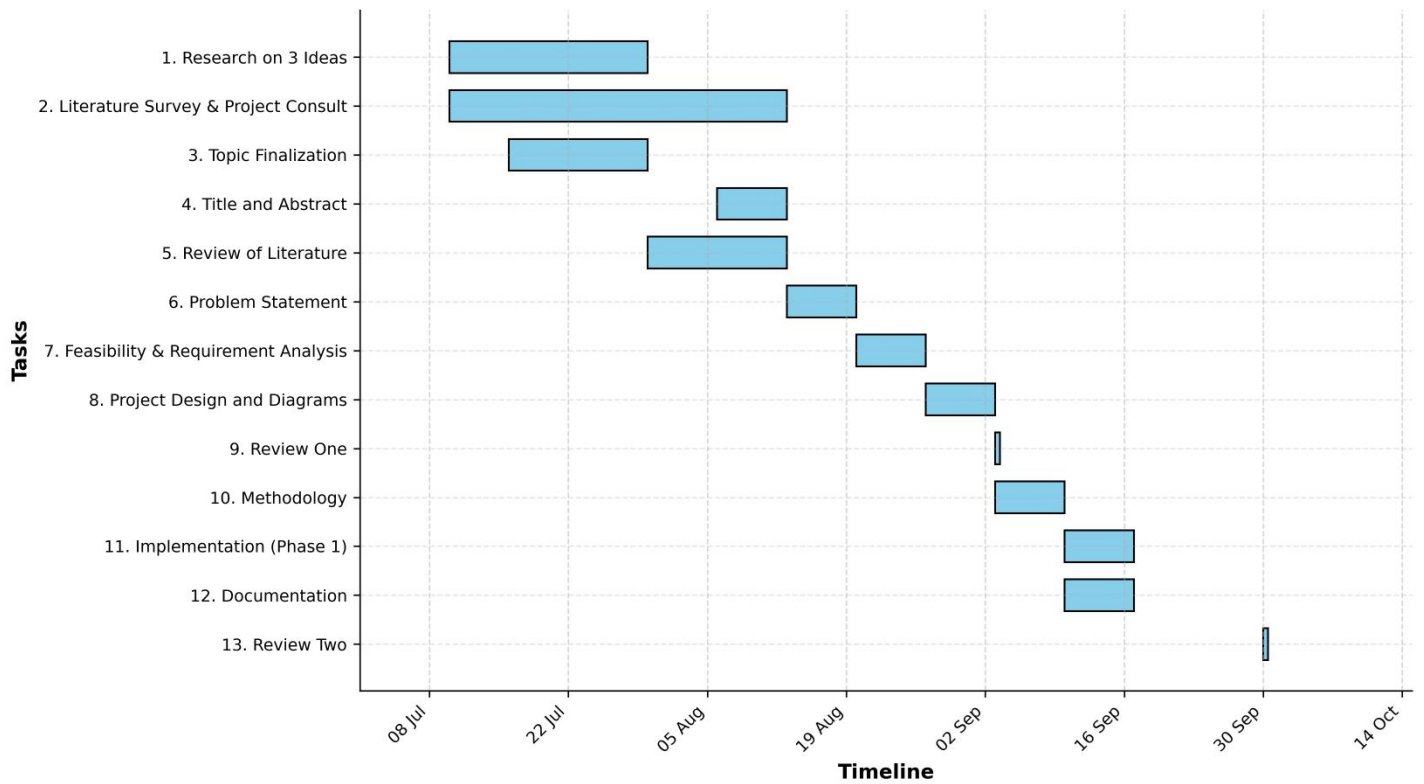
6. Compliance-Oriented Reporting

- Automated reports aligned with GHG Protocol reduce repetitive manual work.
- Improves efficiency in sustainability audits and regulatory submissions.

3.5 Implementation Plan

Sr.No	Tasks	Assigned To	Start Date	End Date	Status
1	Research on 3 Ideas	All Members	10/07/2025	30/07/2025	Complete
2	Literature Survey and Project Consultation	All Members	10/07/2025	13/08/2025	Complete
3	Topic Finalization	All Members	16/07/2025	30/07/2025	Complete
4	Title and Abstract	Ayush Chaurasia, Karan Yadav	06/08/2025	13/08/2025	Complete
5	Review of Literature	Bhakti Ambrale, Shravani Sarang	30/07/2025	13/08/2025	Complete
6	Problem Statement	Ayush Chaurasia, Karan Yadav	13/08/2025	20/08/2025	Complete
7	Feasibility study and Requirement analysis	Bhakti Ambrale, Shravani Sarang	20/08/2025	27/08/2025	Complete
8	Review One	All Members	27/08/2025	03/09/2025	Complete
9	Project design and Diagrams	Bhakti Ambrale, Shravani Sarang	03/09/2025	03/09/2025	Complete
10	Methodology	Ayush Chaurasia, Karan Yadav	03/09/2025	10/09/2025	Complete
11	Implementation	All Members	10/09/2025	17/09/2025	In Progress
12	Review Two	All Members	10/09/2025	17/09/2025	Complete
13	Documentation	All Members	30/09/2025	30/09/2025	Complete
14	Implementation (Phase 2)	All Members	07/01/2026	17/01/2026	Complete
15	Module Integration	Bhakti Ambrale, Shravani Sarang	17/01/2026	30/01/2026	Complete
16	Backend & Database Finalization	Ayush Chaurasia, Karan Yadav	17/01/2026	30/01/2026	Complete
17	Unit Testing and Debugging	All Members	30/01/2026	04/02/2026	Complete
18	Review Three	All Members	07/02/2026	07/02/2026	Complete
19	System Testing and Validation	Bhakti Ambrale, Shravani Sarang	04/02/2026	18/02/2026	Complete
20	Result Analysis and Output Generation	Ayush Chaurasia, Karan Yadav	18/02/2026	25/02/2026	Complete
21	Final Documentation	All Members	25/02/2026	18/03/2026	In Progress
22	Final Review / Project Demonstration	All Members	20/03/2026	20/03/2026	In Progress
23	Final Report Submission and Presentation Preparation	All Members	20/03/2026	End of April 2026	In Progress

3.5.1 GANTT CHART



5.Conclusion

EcoTrace addresses the need for a simple, intelligent, and cost-effective carbon accounting solution for a college campus. By leveraging technologies such as OCR, NLP, and Machine Learning, the system automates the collection, extraction, and analysis of carbon-related data from campus activities, energy bills, and other operational documents, reducing errors and eliminating manual reporting complexity.

The platform enables accurate measurement and tracking of greenhouse gas emissions, identifies carbon hotspots, and generates actionable insights and reports for sustainability planning. Its user-friendly interface ensures accessibility for students, staff, and administrators, allowing the campus community to make informed decisions without specialized knowledge.

In summary, EcoTrace helps the college campus monitor and reduce its carbon footprint, promotes sustainable practices, and contributes to a greener, more environmentally responsible campus while simplifying the carbon accounting process.

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